

Notifications & Push



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In-app inbox, FCM push and live-call ring screen with a clean token lifecycle

Category: **Engagement** | Channels: **FCM + Local** | Status: **Production**

Firestore FCM Push	4 payloads Notif types	Login / refresh / logout Token lifecycle	Yes Local fallback
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1. What it is

WaveMart's notification system combines an in-app inbox with Firebase Cloud Messaging. The inbox is a paginated list of server-side events (price drops, lead updates, system messages). FCM is used for two purposes: (a) to keep the app's providers fresh in the background and (b) to deliver push notifications, including the full-screen ring screen for incoming video calls.

2. How it works

01 Register 1	On login, the device FCM token is POSTed to <code>/api/fcm/register</code> .
02 Foreground 2	<code>FirebaseMessaging.onMessage</code> handler invalidates the relevant Riverpod providers and shows a local notification.
03 Background tap 3	<code>onMessageOpenedApp</code> routes the user to the right screen (chat, call, listing).
04 Inbox 4	<code>GET /api/notifications</code> paginates server-side events; unread count is read via <code>/unread-count</code> .
05 Mark read 5	<code>GET /api/notifications/{id}/read</code> or <code>POST /mark-all-read</code> .
06 Logout 6	FCM token is deleted server-side so the device stops receiving pushes for the logged-out user.

3. Capabilities

Capability	What it does
In-app inbox	Paginated list with filter (all / unread) and bulk 'mark all read'.
Unread badge	Live unread count exposed to all relevant providers.
Push notifications	Foreground and background delivery via FCM.

Per-type payloads	message, incoming_call, lead_update, system - each triggers a different UI flow.
Smart suppression	If the user is already in the relevant chat, the push is suppressed to avoid duplicates.
Lifecycle hygiene	Token deleted on logout; new token generated on next login (stops cross-account leakage).
Settings UI	Read state, swipe-to-delete, empty states and skeletons.

4. Investor relevance

WHY IT MATTERS

Push drives retention. With a clean token lifecycle and per-type payloads, WaveMart can run re-engagement campaigns, price-drop alerts and seller lead updates with negligible cost.

5. Tech surface

Component	Location
FCM service	lib/data/services/fcm_service.dart
FCM API service	lib/data/services/fcm_api_service.dart
Local notifications	lib/core/network/local_notification_service.dart
Inbox screen	lib/presentation/screens/notifications/notifications_screen.dart
Backend service	WaveMart/app/Services/FcmService.php AdminNotificationService.php +